FRESHMAN PROGRAM

RHU offers a freshman program for students who have not earned a Lebanese Baccalaureate or its equivalent, serving as a bridge to eligibility for a bachelor's degree. The freshman program covers two tracks: the Freshman Arts track and the Freshman Science track. Each track encompasses 30 credits that may be completed in two semesters. The required subjects and corresponding credit hours are summarized in the table below.

Program	Humanities/ Social Sciences	Math	Natural Sciences	Free Elective	English	Arabic
Freshman Arts	12	3	3	6	3	3
Freshman Sciences	6	6	9	3	3	3

All students whose native language is Arabic must take an Arabic language course. Students exempted from the Arabic language requirement must take a substitute course to fulfill the 30-credit requirement.

Admission Requirements

To be eligible to join the Freshman Program, an applicant must hold one of the following certificates:

1- A high school certificate based on 12 years of schooling, starting with Grade One, awarded by a government or private school and recognized by the Lebanese Ministry of Education and Higher Education (MoEHE). High school certificates based on less than 12 years of schooling, starting with grade one, are not recognized.
2- A General Certificate of Education (GCE). This applies to applicants with an A-level certificate who have completed a minimum of five subjects at the Ordinary Level and one subject at the Advanced Level or two Advanced Supplementary Levels (AS), excluding languages.

The Lebanese Ministry of Higher Education announced the following requirements to join the Freshman Program at any institute of higher education:

a. Certified copy of the permission from the Lebanese Ministry of Education Equivalence Committee to join the Foreign Program/Freshman class.
b. The following scores are required in one of the aptitude tests below before they can enroll at RHU:

Aptitude Test	Freshman Arts	Freshman Sciences
SAT-Scholastic Aptitude Test	870	950
ACT - American College Test	15	17

Requirements of the Freshman Program Equivalency

1. **If your school is accredited by one of the following accrediting boards,** then the above requirements are enough to grant you admission to the sophomore year after your Freshman Program, and to get the equivalency from the Lebanese Ministry of Higher Education

   Accrediting Boards:

   - Western Association of Schools and Colleges (WASC)
   - Southern Association of Colleges and Schools (SACS)
   - AdvancED (Cognia)
   - New England Association of Schools and Colleges (NEASC)
   - North Central Association of Schools and Colleges (NCASC)
   - Northwestern Association of Colleges and Schools (NACS)
   - Council of International Schools

2. **If your school is not accredited by one of the above accrediting boards,** then you need, in addition to the above two requirements (a & b), to submit:

   ACT Subject Tests during your first year at RHU, before your sophomore year begins.

   Required subjects in the ACT Subject Tests exam are:

   - Mathematics I and any other two subjects for Freshman Arts
   - Mathematics II and two of the following subjects: Biology, Chemistry, and Physics for Freshman Science

   Required scores in the ACT Subject Test:

   - A minimum total score of 51 for Freshman Arts with a minimum score of 15 in each subject
   - A minimum total score of 60 for Freshman Sciences, with a minimum score of 17 in each subject

Please note the following:

Lebanese students cannot declare a major and join the sophomore class until they satisfy the requirements of the Lebanese Ministry of Education and Higher Education and the RHU freshman program.

Study Plan

The following study plan summarizes the courses and credit distribution for the freshman program. Freshman courses are identified by having a “1” as the first digit of the course number.

Freshman Arts

Course No.	Course Name	Credits	Prerequisite
Fall Semester (15 Credits)			
BECN 190	Introduction to Economics*	3	
ENGL 101	Freshman English	3	
MATH 189	Fundamentals of Algebra	3	
	Free Elective	3	
	Humanities/ Social Science Elective	3	
Spring Semester (15 Credits)			
ARAB 180	Arabic Reading and Writing	3	
	Humanities/ Social Science Elective	3	
	Humanities/ Social Science Elective	3	
	Natural Science Elective	3	
	Free Elective	3	

* Required for application to the College of Business.

Freshman Science

Course No.	Course Title	Credits	Prerequisite
Fall Semester (15 Credits)			
ENGL 101	Freshman English	3	
MATH 190	Calculus I	3	
PHYS 190	General Physics I	3	
	Humanities/Social Science Elective I	3	
	Free Elective	3	
Spring Semester (15 Credits)			
ARAB 180	Arabic Reading and Writing	3	
MATH 191	Calculus II	3	MATH 190
PHYS 191	General Physics II	3	PHYS 190
	Natural Science Elective	3	
	Humanities/Social Science Elective II	3	

Courses Description

I. Mandatory Courses

Descriptions of the program's mandatory courses are given below.

ARAB 180	Arabic Reading and Writing	3(3,0)
This freshman-level course is designed for native speakers of Arabic to develop their reading and writing skills. This course is conducted in Arabic. Students will read and discuss essays, articles, and other readings, and write essays. It fulfills the Freshman Arabic language requirement. The course is mandatory for all native Arabic speakers.		

BECN 190	Introduction to Economics	3(3,0)
This course is designed to help students develop a basic general understanding of economic principles, issues, history, and institutions. It develops their economic literacy and teaches them how economics relates to the everyday lives of individuals, businesses, and society. Topics include markets, firms, economic systems, the role of government in the economy, capitalism, macroeconomic indicators, and the basics of supply and demand.		

ENGL 101	Freshman English	3(3,0)
This course is designed to help students develop their reading and writing skills. It focuses on the interrelatedness between reading and writing, whereby students critically read and analyze passages in different rhetorical modes (classification, cause/effect analysis, and comparison/contrast analysis) and learn to write similar essays in these modes. Some grammatical and structural elements are reviewed to help students improve the accuracy of their writing. In this course, students are also introduced to research skills such as evaluating sources, paraphrasing, summarizing, and quoting while following the APA style.		

MATH 189	Fundamentals of Algebra	3(3,0)
Real number systems, radicals and rational exponents, polynomials, factoring, fractional expressions, lines in the plane, functions and their graphs, inverse functions, solving equations and inequalities, real zeros and the fundamental theorem of Algebra, exponential functions and their graphs, logarithmic functions and their graphs.		

MATH 190	Calculus I	3(3,0)
Functions and their graphs, limits and continuity, differentiation, applications of derivatives, Extreme values and mean value theorem, definite and indefinite integrals, and the substitution method.		

MATH 191	Calculus II	3(3,0)
Covers limits and indeterminate forms with L'Hôpital's Rule, optimization, antiderivatives, and the Fundamental Theorem of Calculus. Topics include substitution methods, areas, volumes (cross-sections, shells), arc length, and surfaces of revolution. It introduces parametric and vector functions with applications to motion, arc length in space, curvature, and normal vectors. **Prerequisite**: Math 190.		

PHYS 190	General Physics – Mechanics	3(3,0)
This course introduces students to the basic concepts and principles of Mechanics: Vectors, displacement, velocity, acceleration, motion in 1D, projectile motion, forces, Newton's laws of motion, work, energy, momentum, collision, torque, and rotational motion.		

PHYS 191	General Physics - Electricity and Magnetism	3(3,0)
This course is an introduction to electricity and magnetism. In this course, students will explore electric charges, electric forces and fields, electrical energy and potential, capacitance, direct-current circuits, magnetic force, magnetic field, induced voltage and inductance, alternating-current circuits, and electromagnetic waves. **Prerequisite**: PHYS 190.		

II. Electives

Descriptions of the possible elective courses are given below.

BACC 150	Introduction to Accounting	3(3,0)
Students in this course will learn the basic concepts and techniques of accounting, including accounting information systems, costing, estimation, and some aspects of management accounting.		

BADM 155	Introduction to Law	3(3,0)
This course covers the basic tenets of the legal system and how they may apply to various business sectors and industries.		

BIOL 102	Introduction to Biology	3(3,0)
This is an introductory course to the fundamental principles of biology that covers the chemical basis of life, the structure and function of cells and systems, genetics, and human body systems.		

BIOL 170	Basic Concepts in Biology	3(3,0)
A freshman-level course that introduces students to the basic concepts of genetics, evolution, and ecology and explores the forms and functions of plants and animals.		

BMGT 150	Basics of Management	3(3,0)
This course provides a general overview of management basics, including principles of management science, buying and selling, and management.		
BMGT 155	**Introduction to Hospitality Management**	**3(3,0)**
Students will learn the basics of different aspects of hospitality, such as food, lodging, beverages, and pastries. They will also be exposed to hotel, restaurant, and resort hospitality operations.		

BMKT 150	Introduction to Business and Commerce	3(3,0)
This course introduces students to the nature of business, the types of commercial activities, market analysis, and the interrelationships among commercial activities.		

CCEE 111	Computer Programming	3(3,0)
This course introduces students to fundamental programming concepts such as variables, loops, conditional statements, and event handling. It also helps students use math and computer code to think creatively.		

CCEE 112	Internet Development and Support	3(3,0)
This course covers issues related to the development and implementation of internet-related tools and services. Topics include Internet organization, site registration, e-mail servers, Web servers, Web page development, legal issues, firewalls, multimedia, TCP/IP, service providers, FTP, list servers, and gateways.		

CCEE 121	Computer Technology	3(3,0)
This course is designed to help students understand the basic functions of a computer as a human-machine interacting **system**. It covers the basics of computer technology in today's world. Students will focus on the following components of computers: keyboards, operating systems, word processing, spreadsheets, database management, presentation graphics, and the internet.		

CCEE 122	Computer Hardware	3(3,0)
This course introduces the technical aspects of computer hardware units. Topics covered are computer organization, communication inside the computer, and communication with the outside world.		

CCEE 123	Introduction to Computer Troubleshooting	3(3,0)
In this course, students will learn how to set up a computer, diagnose simple faults, and troubleshoot them.		

CCEE 131	Introduction to Audio-Visual Technology	3(3,0)
In this course, the students will be introduced to the technologies involved in the Audio and Video Communication systems. The course includes video production, multimedia, sound, animation, and photography.		

CCEE 141	Switching Systems Technology	3(3,0)
This course includes the public switched telephone network; the telephone and the local loop architecture; inter-exchange networks and signaling; the evolution of switching technology and architectures, and a comparison of various systems; and traffic statistics and the theory of space-division and time-division switching networks.		

CCEE 142	Basics of Analog and Digital Communications	3(3,0)
This course covers analog and digital communication systems: basic principles of telecommunication systems, operation and application of various transducers, and basic principles of modulation and demodulation.		

CHEM 160	Chemistry I	3(3,0)
Students study the principles and methods of chemistry, including atomic structure, periodic trends, writing formulas, and determining the names of both inorganic and organic formulas.		

CHEM 170	Chemistry II	3(3,0)
Students learn about acids, bases, and chemical equilibria; the groups in the periodic table, organic chemistry, and nuclear chemistry will also be surveyed.		

CIVE 105	History of Architectural Technology	3(3,0)
The study of architectural technology from the Greek civilization to the present stresses the development of structural systems and the exploration of materials. This course will explore the interaction of building design and historic socio-economic determinants.		

CIVE 120	Introductory Strength of Materials	3(3,0)
The study of stress and strain, deformation, simplified analysis of torsion, shear, and stresses in beams, beam design, beam deflection, and columns. Reference to applications for civil and mechanical engineering technology, introduction to materials testing.		

CIVE 121	Concrete Construction Methods and Concrete Structures	3(3,0)
Emphasis is placed on applying structural design principles for new and existing concrete structures. Study of concrete structures using basic physical laws. The course covers the basics, from foundation to roof, and the properties and applications of concrete, emphasizing concrete, masonry, and shallow foundation systems. Conditions encountered during renovations and their solutions will be included.		

CIVE 135	Plumbing and Air Conditioning	3(3,0)
Review of water distribution plumbing and related fundamentals. This course covers the basics of piping, valves, backflow prevention, water quality, and hydraulic fundamentals. Review of air-conditioning fundamentals, piping, capacity requirements, and recent developments in materials, fixtures, and appliances.		

CIVE 140	Introduction to Surveying	3(3,0)
A study of topographic surveying and mapping. Determination of land areas, construction surveys and layout, control surveys, boundary surveys, route locations, and street layout. Provides experience with the use of equipment, instruments, and the fundamental techniques of surveying		

CIVE 141	Fundamentals of Topometry	3(3,0)
Measurement and graphic description of topography, topometric descriptions of objects using various techniques in mapping		

CIVE 142	Photogrammetry	3(3,0)
Principles and techniques of vertical aerial photograph photogrammetry, distance, direction, area, and photograph scale calculations and interpretation.		

CIVE 143	Geodesy	3(3,0)
A horizontal and vertical datum, gravitation and centrifugal forces; measurements of gravity and reduction of gravity, geoid, and ellipsoid, deflection of the vertical; orthometric and dynamic heights		

CIVE 144	Mapping	3(3,0)
Topographic mapping, spatial analysis, map production and cartography, spatial reference systems, scale, projections, generalization, and symbolization.		

CIVE 145	Geomorphology	3(3,0)
Surface processes and landforms; their implications for land use and for present, past, and future landscape development; weathering; glacial and fluvial processes; mass wasting; and tectonic controls on landforms.		

CIVE 146	Construction and Surveying Regulations and Laws	3(3,0)
Construction and surveying regulations and laws in Lebanon and their applications to land usage, urban development, and residential and commercial buildings, with emphasis on the construction field, the law of practice, and professional registration		

CIVE 148	Introduction to Public Works	3(3,0)
This course introduces students to construction, alteration, repair, or improvement on public land that is executed at the cost of the state or any other local public agency. Skills in Public administration and supervision of public works are emphasized.		

CVLN 160	Monotheistic Religions: A Comparative Study	3(3,0)
This course covers the three monotheistic religions. It compares the idea of God and the law in Judaism, Christianity, and Islam.		

CVLN 161	The Arabs	3(3,0)
This course reviews the history of the Arabs from the pagan days through the twentieth century.		

CVLN 170	God and Creation: East and West	3(3,0)
A literary attempt to understand the universe's origins as found in texts from various world cultures.		

CVLN 180	Youth and the Growing Up Process	3(3,0)
An introduction to the themes and challenges of autonomy and independence as experienced by youth, studied through major literary works of the past centuries.		